Work Update

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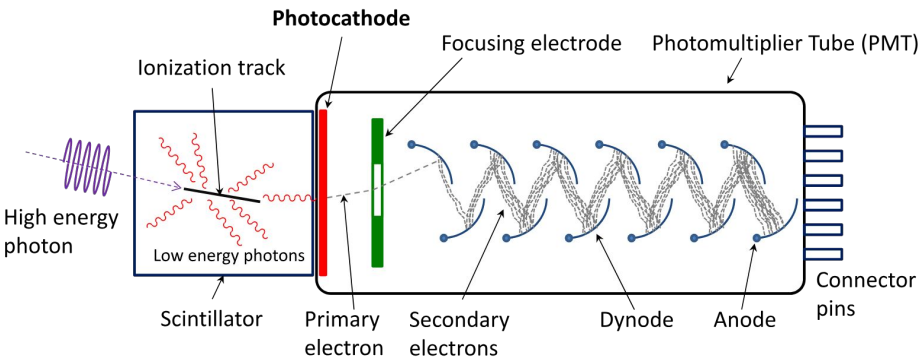
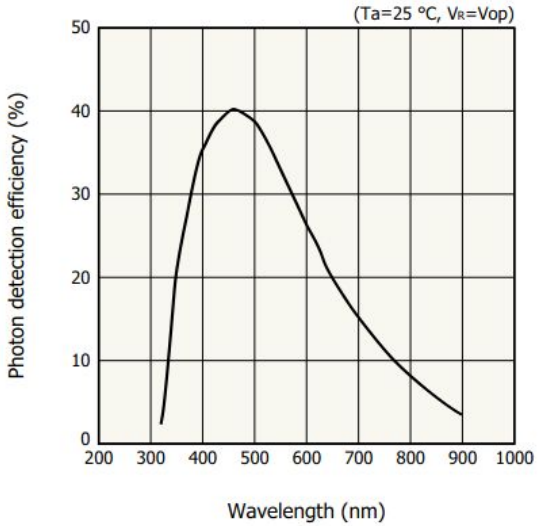
Quick Recap

Unfolding the SiPM Quantum efficiency for the detection of scintillation photons

Introduction

- In Scintillators, optical photons produced are readout at the rear surface by converting them into electrical signal using PMT/ SiPM.
- PMTs / SiPM can process only a certain wavelength of the optical photons, and with a certain probability termed as **Quantum efficiency**.

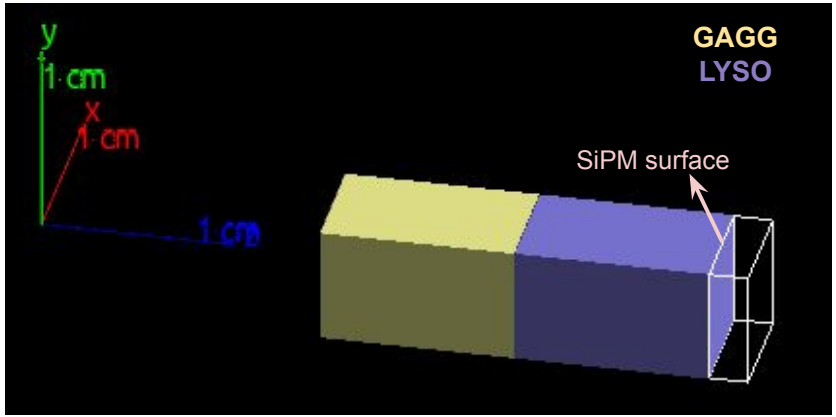


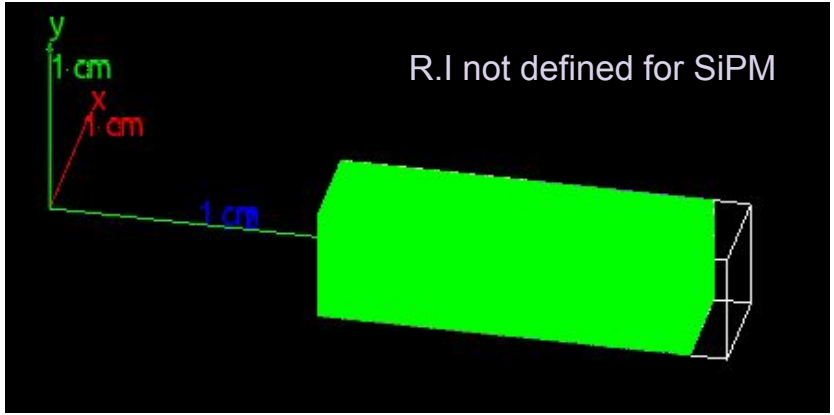


SiPM MPPC ARRAY,
[ModelNo:S13361-3050AE-08-HSN:85414900]

Detector geometry

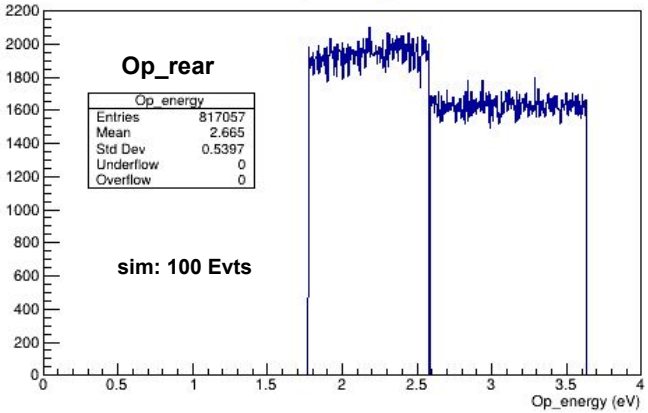
- An stack of GAGG and LYSO scintillaors.
- Dimension: 6mm * 6mm * 10mm.
- Placed a cuboidal volume of with Si having dimension 6mm* 6mm* 2mm at the rear surface of LYSO.
- Optical surface:
  - Between Air and Scintillator boundary
    - Reflectivity 0.99
    - Dielctric_metal
  - Between LYSO and SiPM volume
    - Defing the efficiencies of the surface to detect the Op photons of different energies.
- Shooting 511 keV photons from the left, at the center of GAGG face.

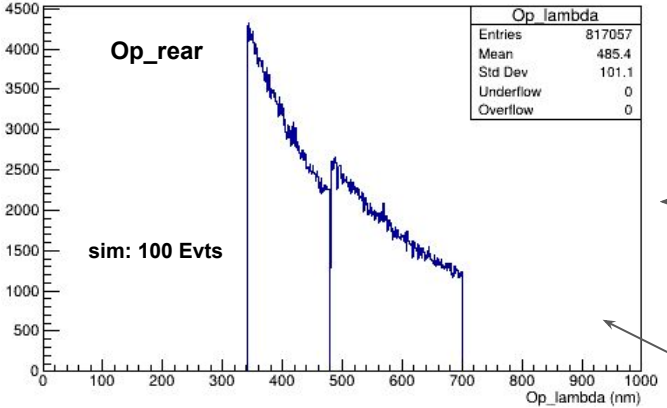




Have referred to G4 example: optical/extended/LXe to apply QE.

Testing whether G4 inbuilt efficiency is working

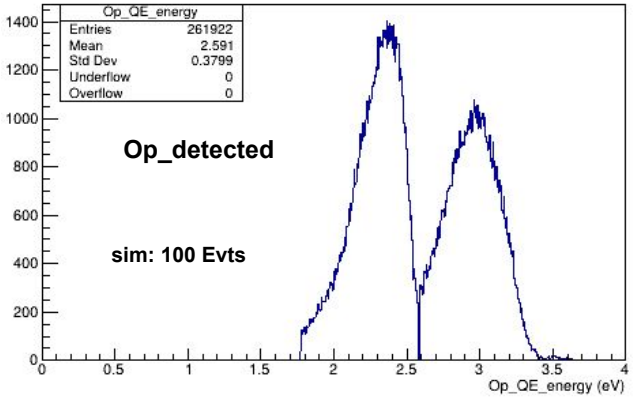


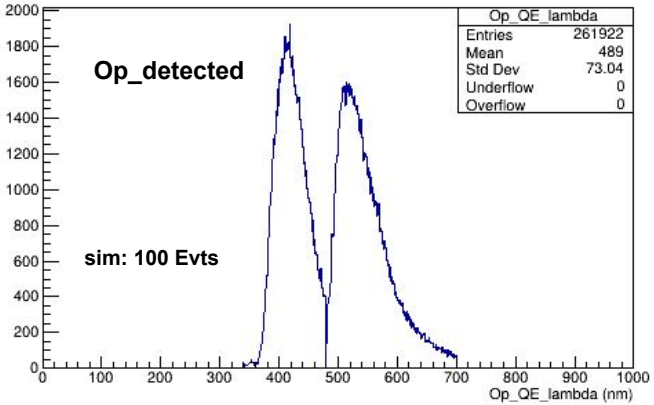


Producing the optical photons such that probability of each Op energy is same.

dN/dE = constant

$dN/d\lambda$ = constant / λ^2

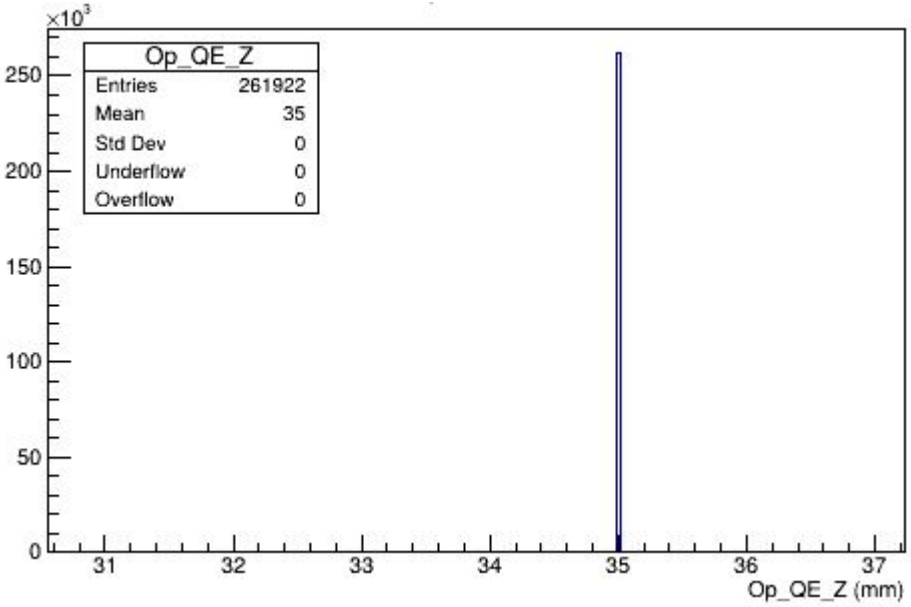




After applying relative fraction QE of the SiPM. The QE is defined to be similar to the emission spectrum of GAGG and LYSO

Built in efficiency in G4 seems to be working.

Detected photon Z



- A clear spike at z = 35 mm makes sense because there is SiPM surface and detection only possible there because only that surface has non zero QE.
- I have made sure that filling the histogram outside OpZ ==35 mm condition.

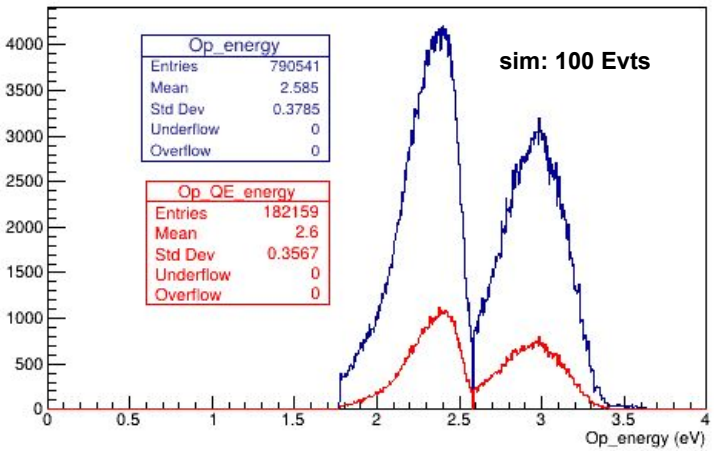
```
  if (nullptr != fBoundary) boundaryStatus = fBoundary->GetStatus();
  if (boundaryStatus == Detection) {   //detection only possible at SiPM surface
    fRunAction->h_Op_QE_lmbda->Fill(Op_lmbda);
    fRunAction->h_Op_QE_energy->Fill(postStepPoint->GetKineticEnergy()*1e6);
    fRunAction->h_Op_QE_Z->Fill(TrkPos[2]);
  }
}
```

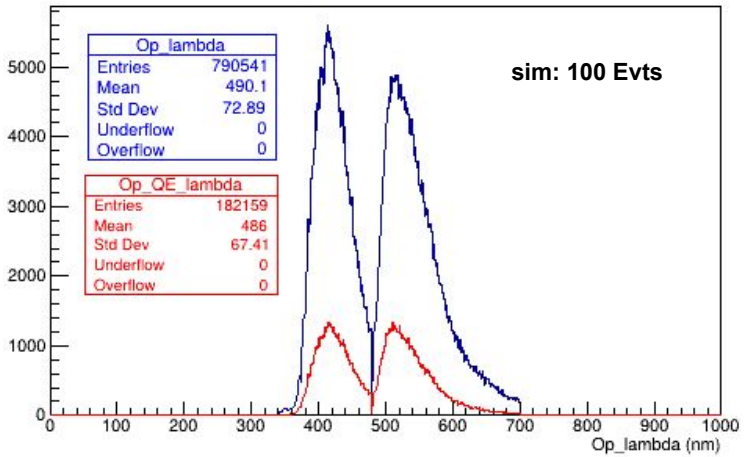
```
//counting the Op photons which gets detected after QE of SiPM
G4OpBoundaryProcessStatus boundaryStatus = Undefined;

auto part = track->GetDefinition();
// find the boundary process only once
if (nullptr == fBoundary) {
  G4ProcessManager* pm = part->GetProcessManager();
  G4int nprocesses = pm->GetProcessListLength();
  G4ProcessVector* pv = pm->GetProcessList();
  for (G4int i = 0; i < nprocesses; ++i) {
    if (nullptr != (*pv)[i] && (*pv)[i]->GetProcessName() == "OpBoundary") {
      fBoundary = dynamic_cast<G4OpBoundaryProcess*>((*pv)[i]);
      break;
    }
  }
}
```

Correction in QE for a wavelength

Earlier I was providing the relative fraction of efficiency, which was not correct in terms of defining Quantum efficiency.

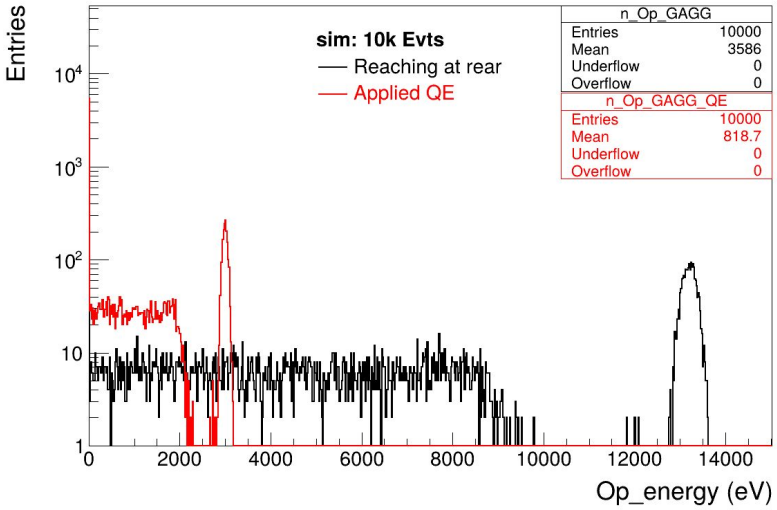
- thats why simulated fraction seemed larger than the expected one.
- Now giving the absolute fraction, there numbers makes sense.

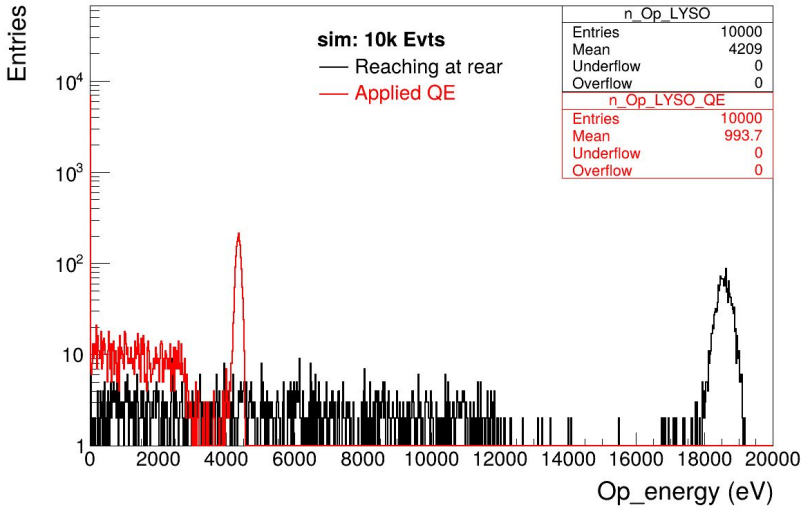




correct QE file

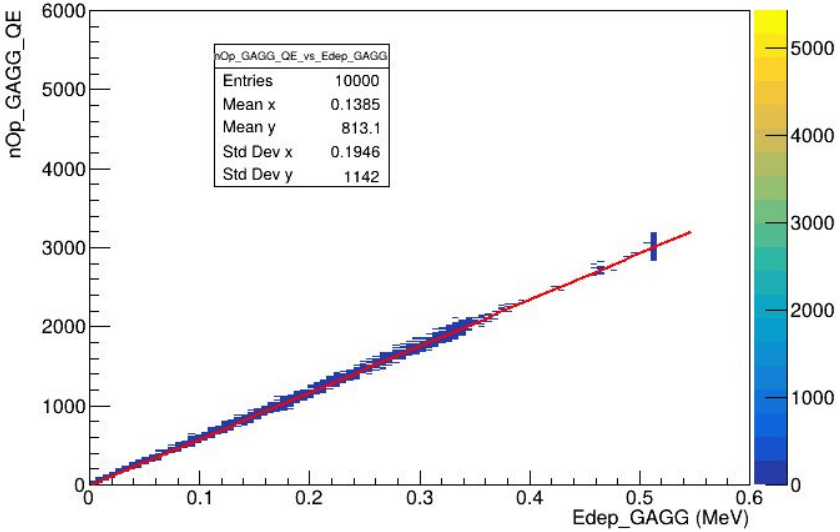
Number of Optical photons detected in GAGG and LYSO



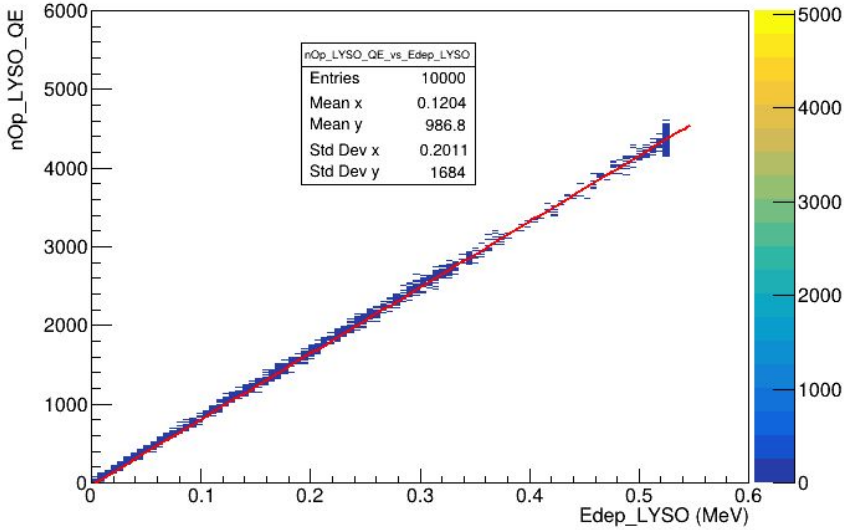


- The number of entries in the partial energy deposits in applied QE case is greater than the one which reaches to PMT. Width at full energy deposit seems to be decreased in QE applied case.

nOp vs Edep after applying QE



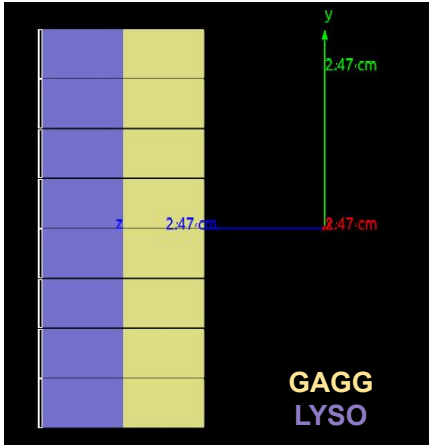

```
*********************************************
Minimizer is Linear / Migrad
Chi2                      =      88031.4
NDf                       =          847
p0                        =     -9.48649   +/-   1.25864
p1                        =      5874.24   +/-   5.25271
```

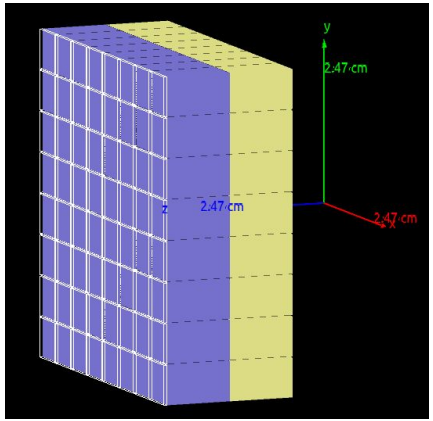


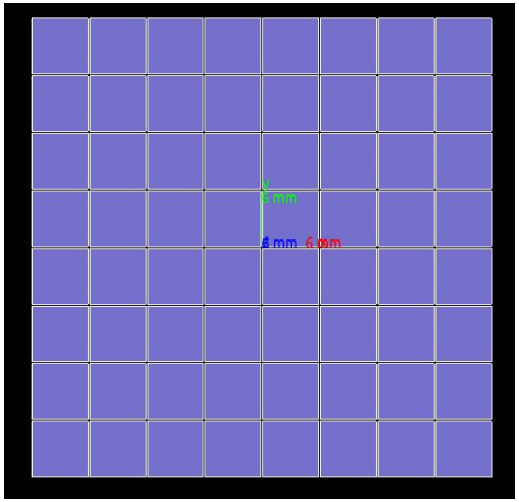

```
Minimizer is Linear / Migrad
Chi2                      =       143432
NDf                       =          856
p0                        =     -29.8553   +/-   1.51411
p1                        =      8373.97   +/-   6.44707
```

Multi Pixelated detector with pixels containing GAGG and LYSO crystals

Detector geometry



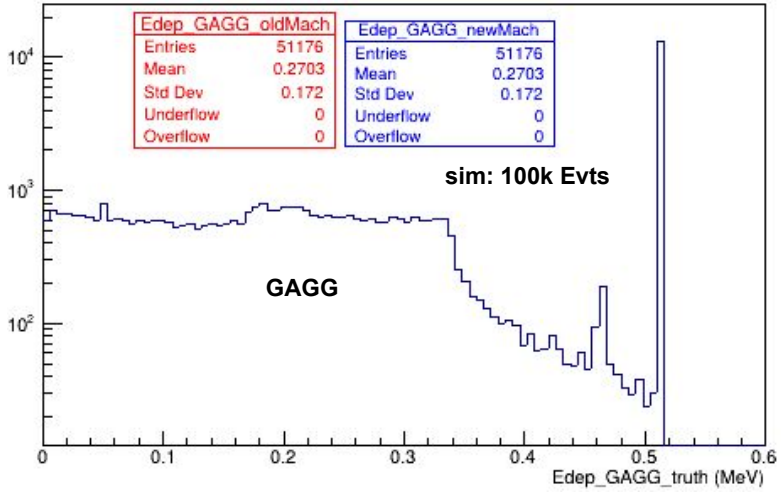


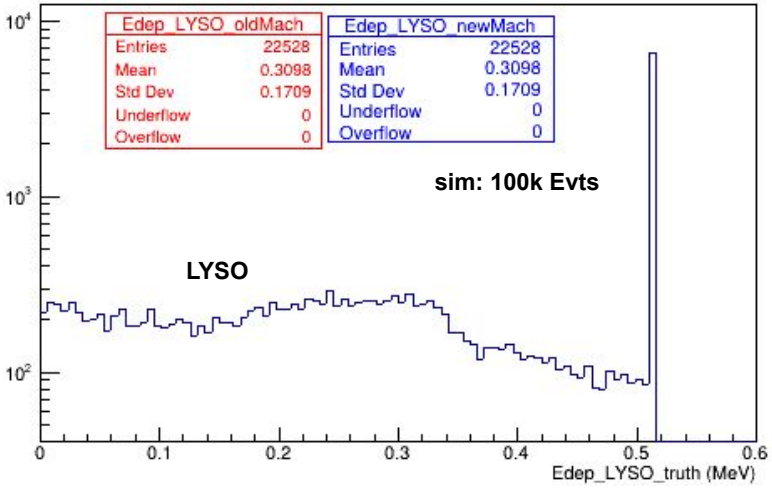


Scint dim: 6mm * 6mm * 10 mm
SiPM dim: 6mm * 6mm * 0.25 mm

For initial checks, not invoking the scintillation in the materials, i.e. working with truth level info only.

True Energy deposited in GAGG and LYSO



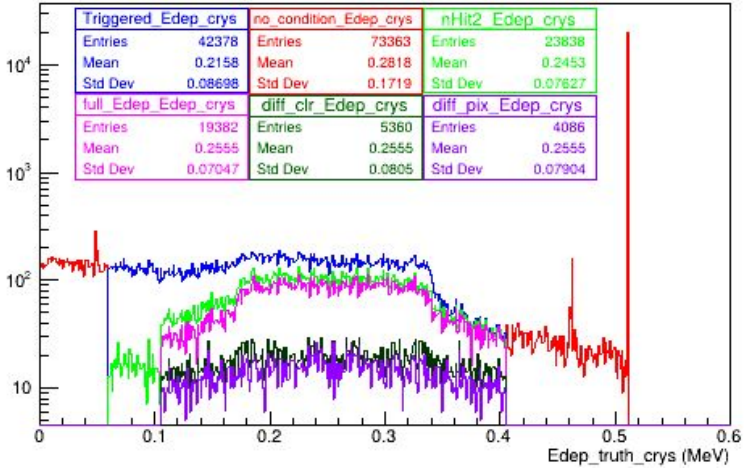


- There was a small bug while initializing a boolean in the new machinery.
- After fixing Edep distribution, both the distributions make more sense.
- There is no two additional peaks in LYSO.
  - That is dependent on how the LYSO material is defined in the simulation.
  - In the general formula of LYSO $Lu_{2(1-x)}Y_{2x}SiO_5$, I used x=0.1 and made it to the nearest integer ($Lu_9YSi_5O_{25}$)
  - In some other composition of LYSO two additional peaks were visible.

Numbers for different event topologies

For a hit to be called triggered: 60keV ≤ Edep ≥ 450 keV
Total triggered: 27.16% of 100k Evts

Category	Percentage out of triggered events
Exactly 2 hits	43.89%
Full Edep in 2 hits	35.69%
In different colours	9.87%
In different pixels	7.52%



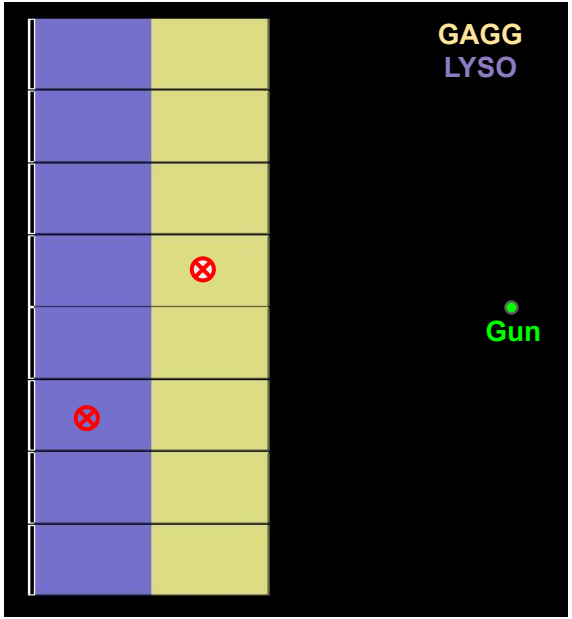
Reconstructing the Compton scattering angle

Event selection

- We select events where:
  - Full energy is deposited in 2 **crystals**.
  - Two hits are present in different colours of the different **pixels**.

Identifying the scatterer (assuming single interaction in scatterer)

- **Hypothesis 1:** hit on GAGG is scatterer
  - Implies that forward Compton scattering
  - hypothesis true if Edep <= 255 keV.
- **Hypothesis 2:** hit on LYSO is scatterer
  - Implies backward Compton scattering
  - Hypothesis true if Edep ∈ (255, 341) keV

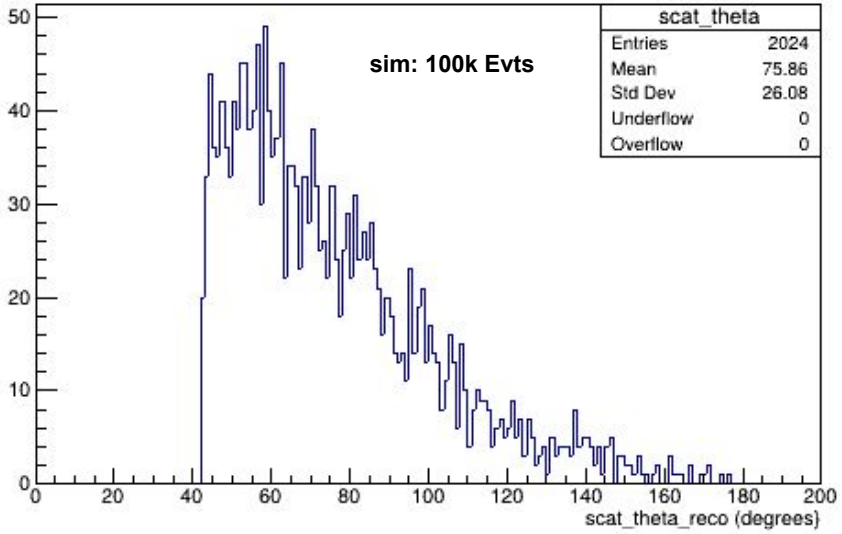


Calculating the scattering angle:

$$E' = E - E_{dep}$$

$$E'(E,\theta) = \frac{E}{1 + \frac{E}{m_e c^2}(1 - cos\theta)},$$

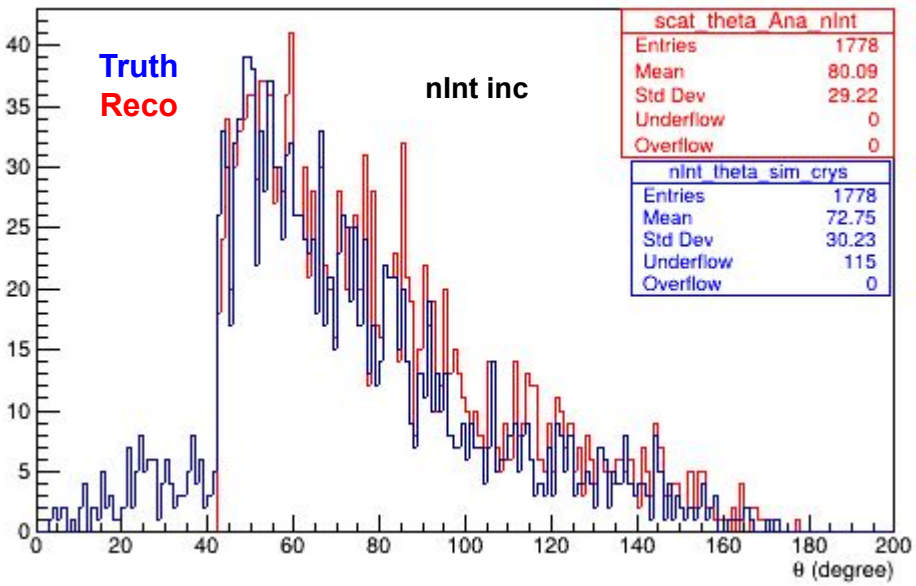
Distribution of reconstructed theta

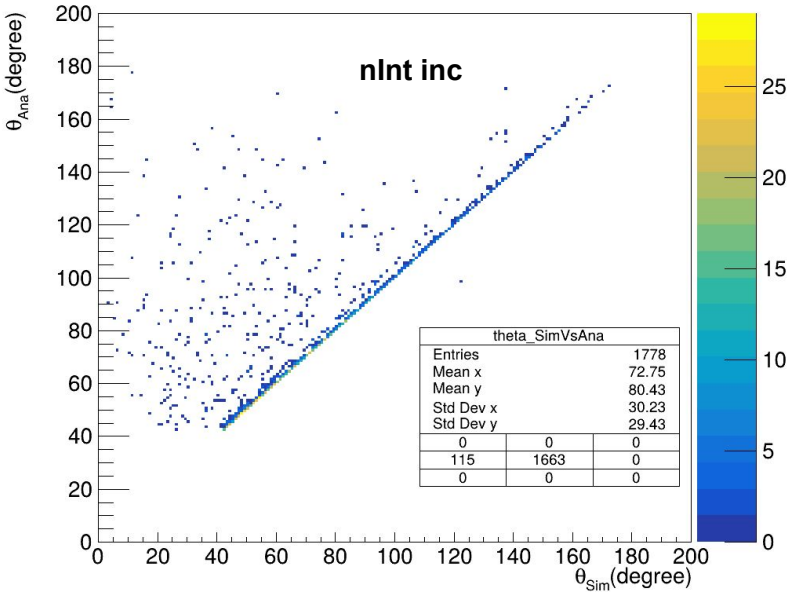


- Doing some checks and comparing with the truth theta is still remaining.

Comparing the Truth scat angle with reco scat angle

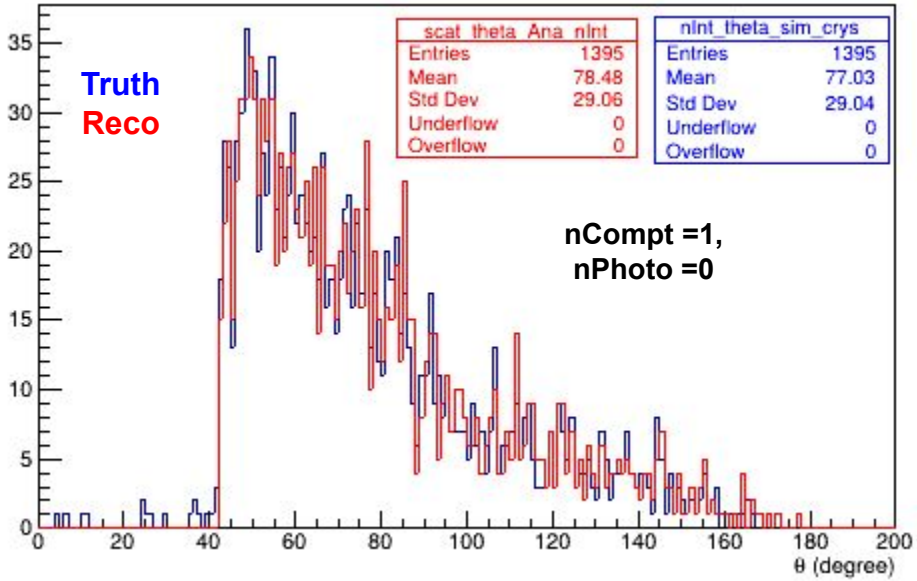
Without putting the condition on number of Compton and photo electric absorption in a crystal

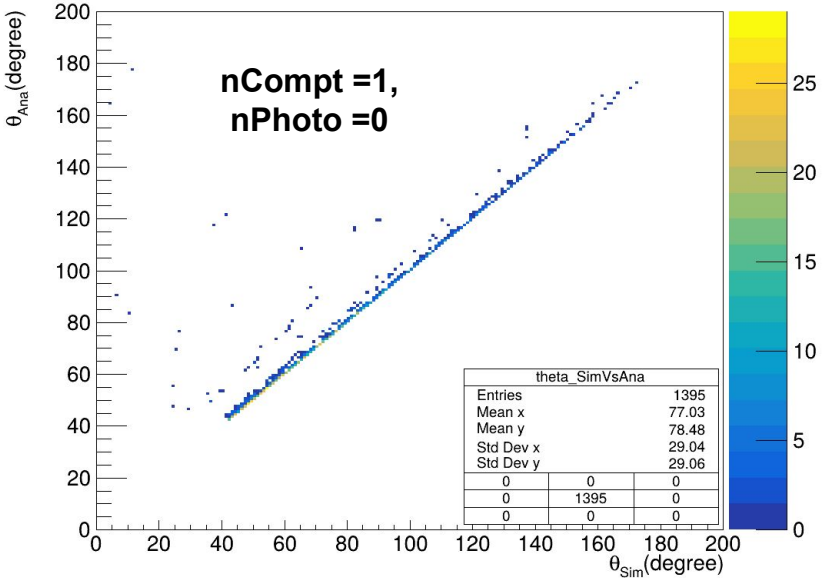




- There is slight change in randomization of gamma direction wr.t to previous slide.
- Event away from the y=x line might correspond to the multiple compton in a single crystal
- In the 2d plot, there is no entry where ana theta is less than sim theta. Not sure why?

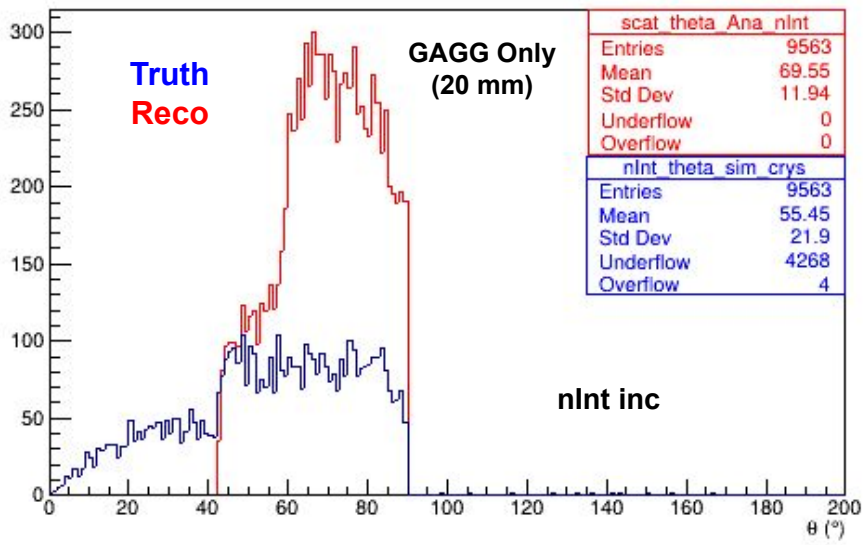
After putting the condition on number of interactions

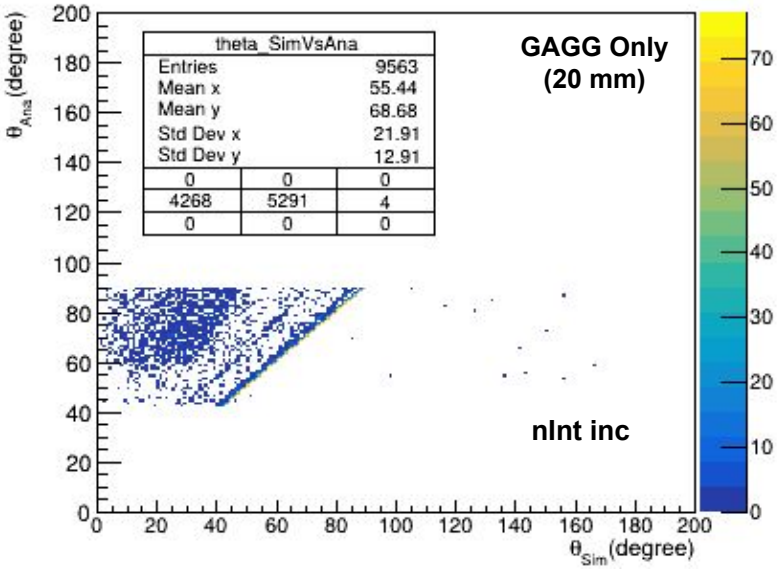




- Have put the condition on the scatterer crystal that there should be only one Compton scattering in the crystal and no photoelectric absorption.
- In reality such information won't be available.

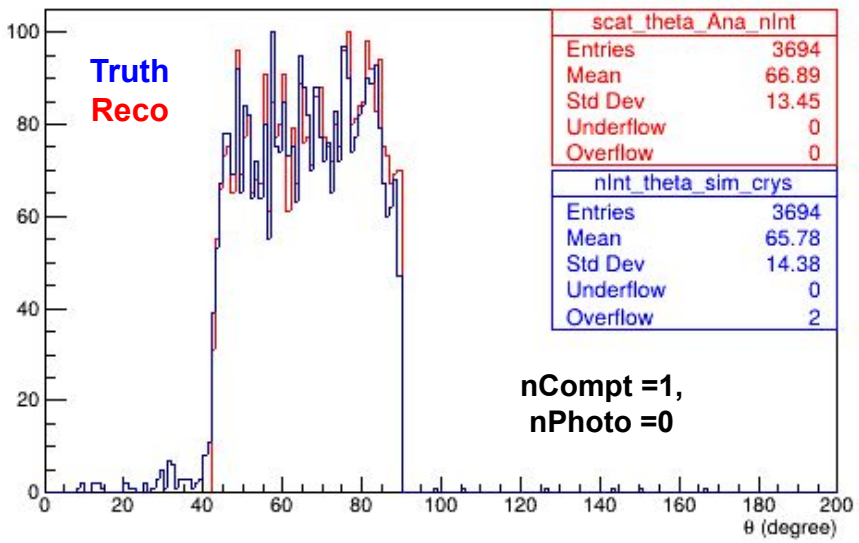
Angle reconstruction without CCal (single crystal)

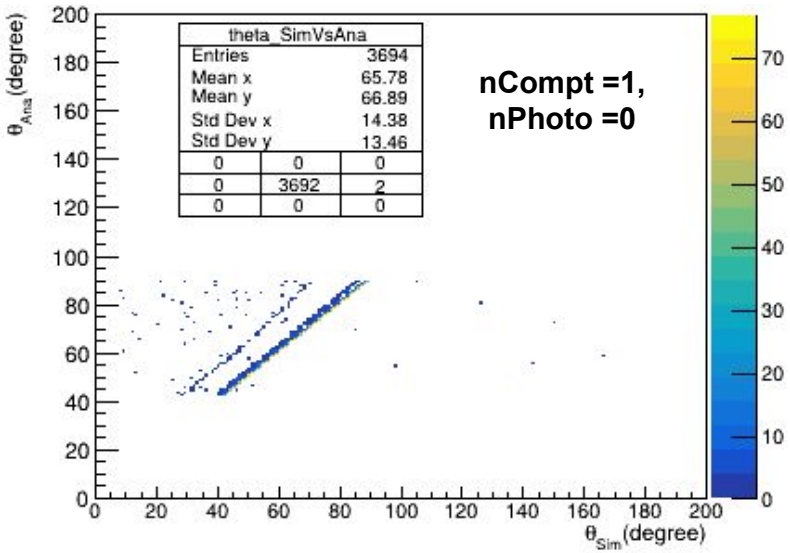




- Underflow in the truth might be corresponding to crystals, when there is photoelectric absorption only happened in the scatterer at truth level.
- Scattering angle reconstruction in Single crystal seems to be worse than expected.

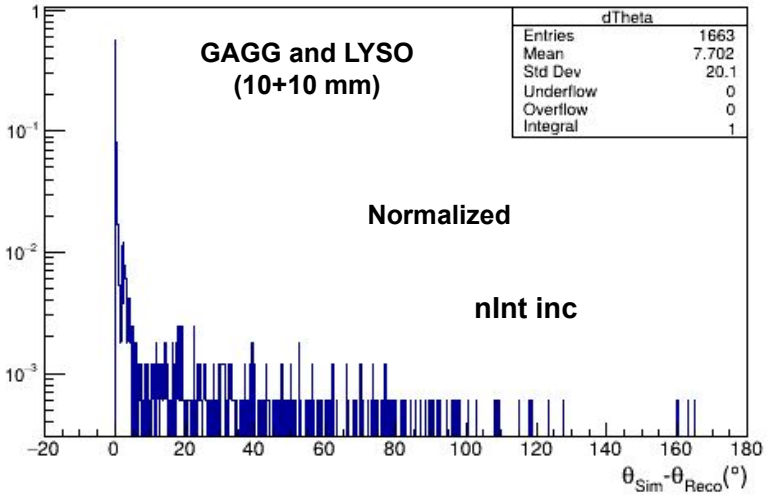
After putting the condition on interactions in scatterer

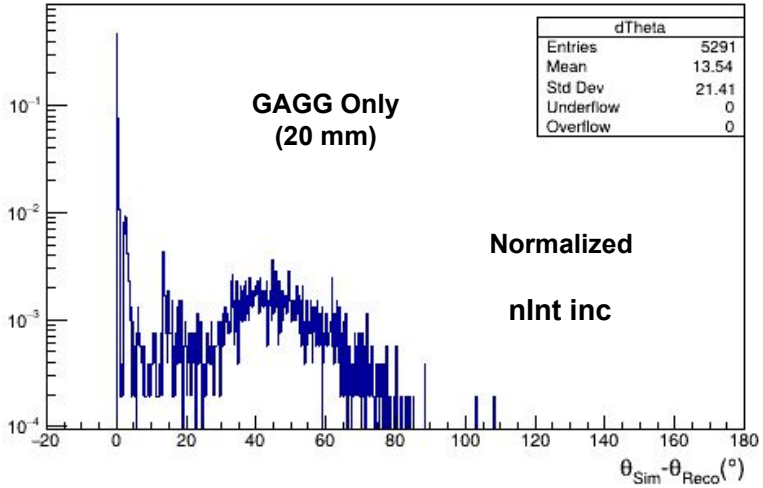




- After putting the conditions on interactions in scatterer angle reconstruction seems to be reasonably good in for the single crystal case.
- It indicates that with CCal setup somehow we have better discrimination of single compton scatterings from the multiple one in the scatterer (other than forward vs backward scattering).
- However reduction in the number of events after applying the above conditions seems to be higher than expected.

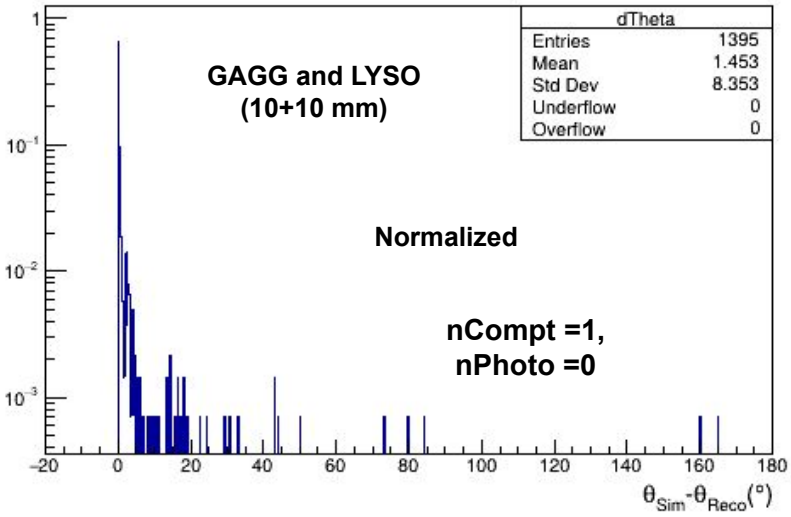
Comparing the theta reconstruction (CCal vs single cryst)

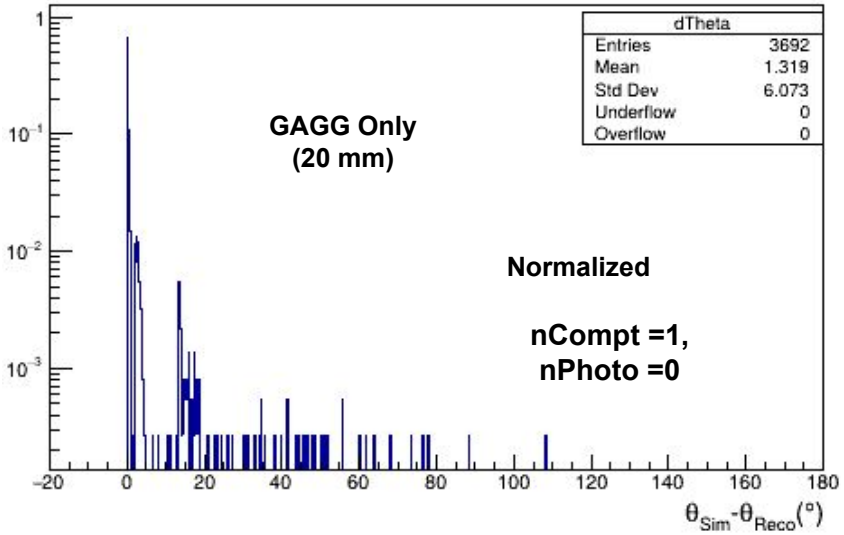




- Filling the absolute value of the difference between truth and reco in the histograms.
- The histogram means is significantly less in CCal setup, Indicating the better scattering angle reconstruction compared to the single crystal setup.

After putting the condition on interactions in scatterer

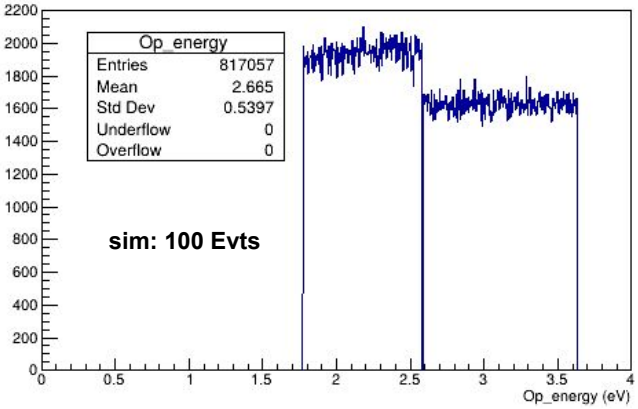


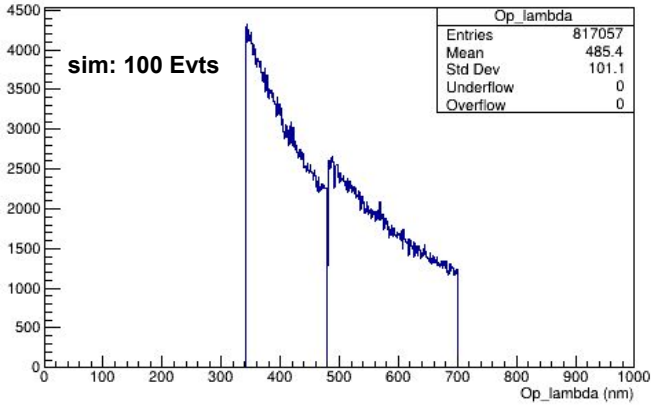


Both reconstructions seem to comparable when there is a condition applied on the interactions in the scatterer.

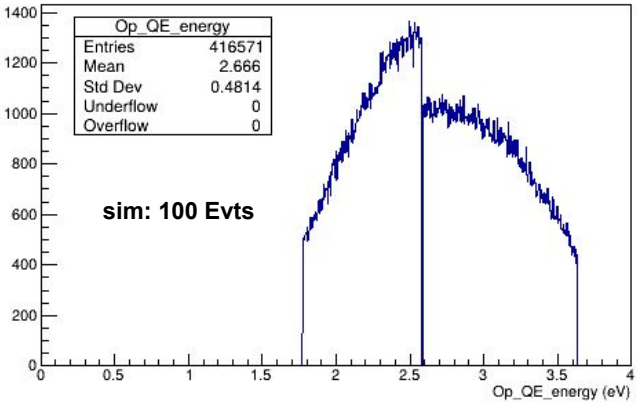
Back Up

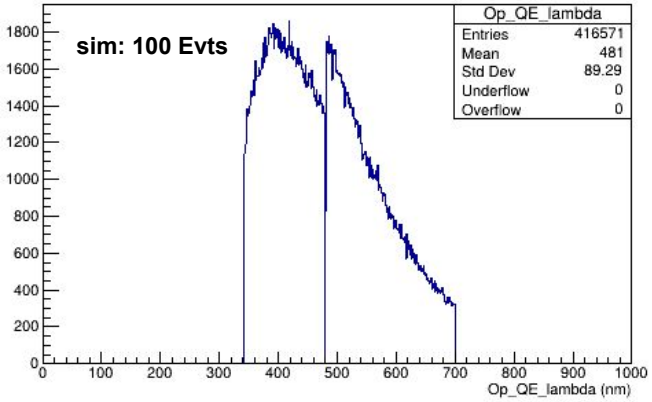
Testing the QE with flat distribution in Op energy





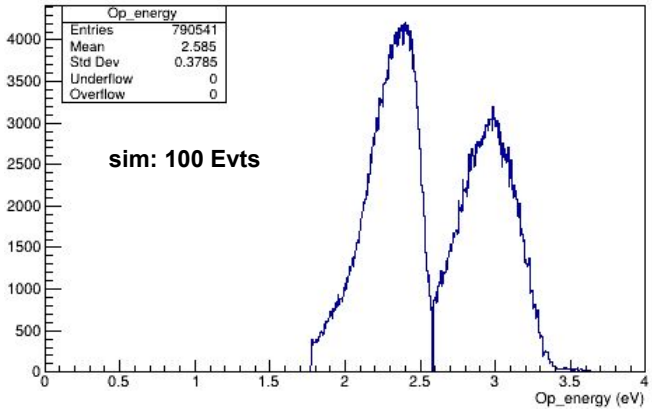
Op reaching to LYSO rear.

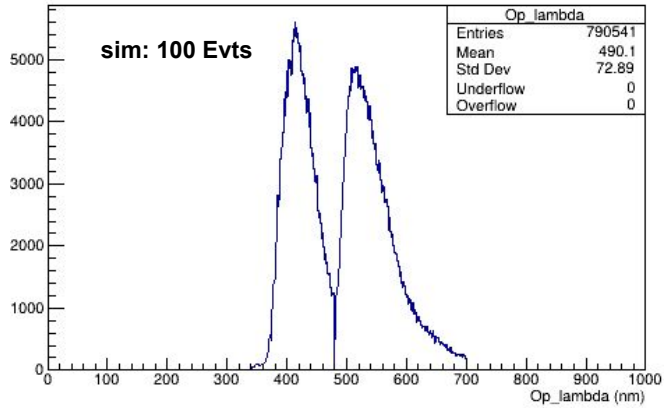




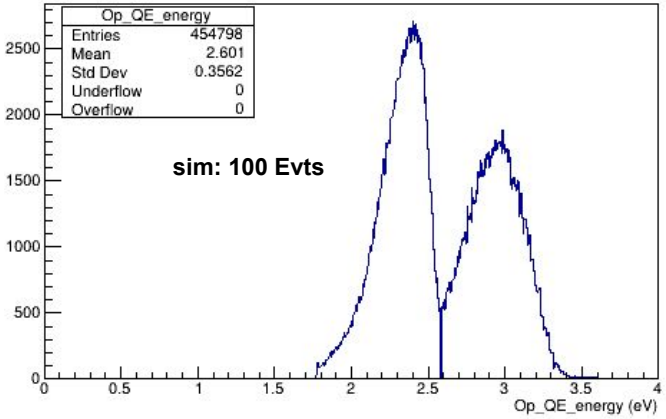
Op detected (QE applied)

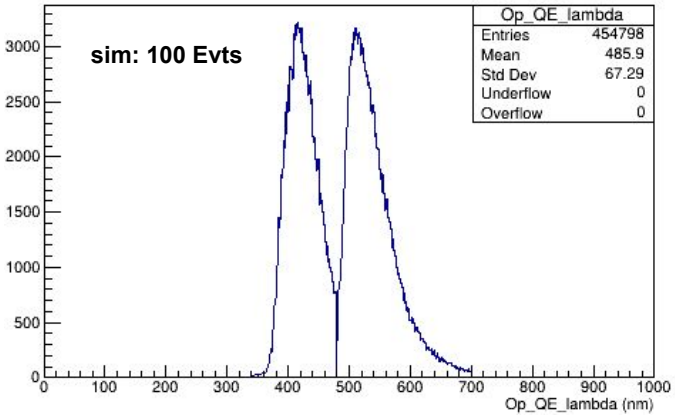
Applying QE in the original GAGG LYSO emission (lambda seperated)





Op reaching to LYSO rear.



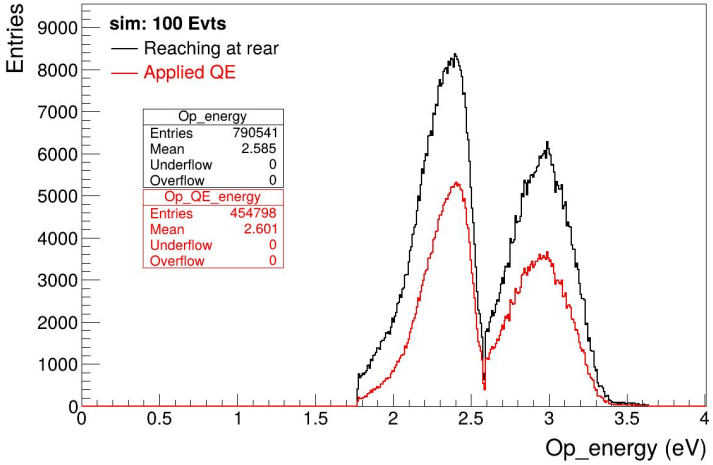


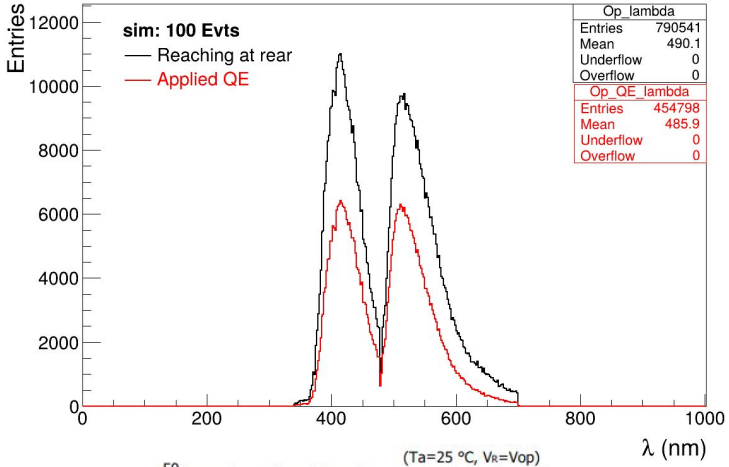
Op detected (QE applied)

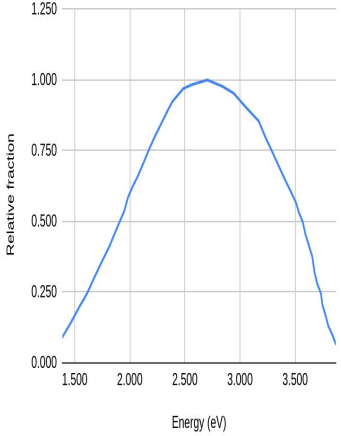
Shape doesn't seem to be changing, but numbers are changing after QE.

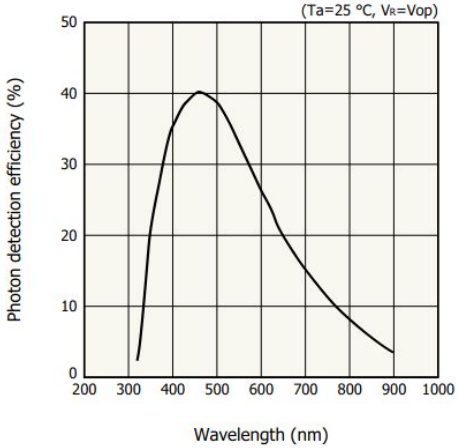
Comparing the emission spectrum after QE

Date: 19/02/26



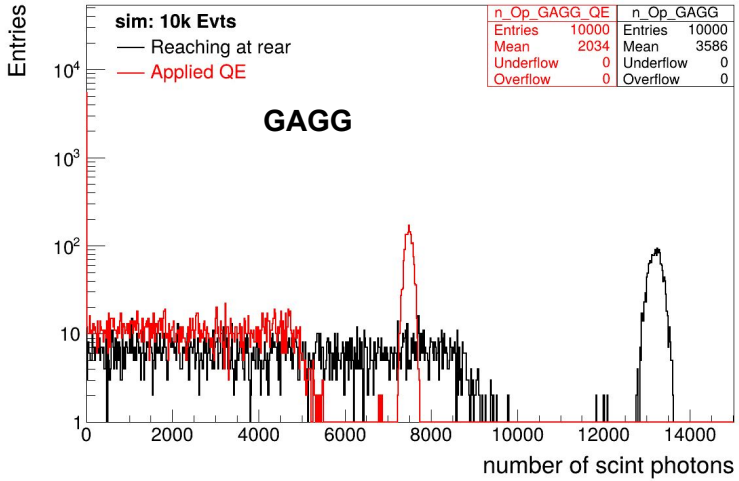


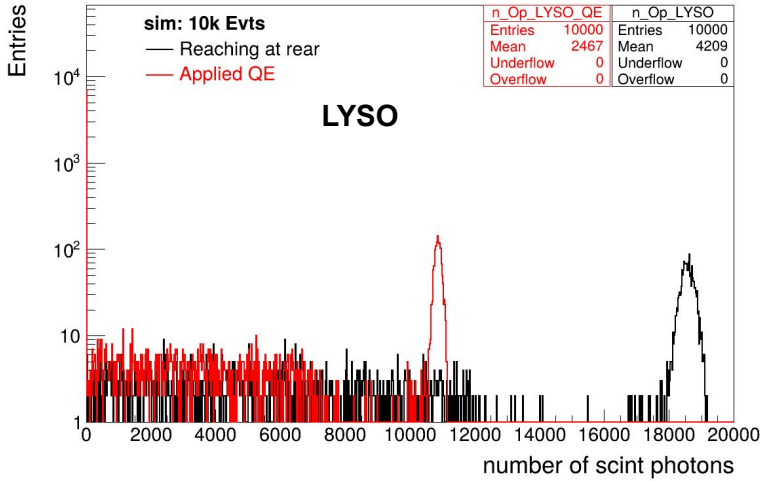




SiPM QE CSV file

Comparing the distribution of nScint photons after QE





- There seems to be a similar reduction in GAGG and LYSO after applying QE.
- Numbers are still larger than expected.

Implementation of surface efficiency in Geant4

```
inline
void G4OpBoundaryProcess::DoAbsorption()
{
              theStatus = Absorption;

              if ( G4BooleanRand(theEfficiency) ) {

                 // EnergyDeposited =/= 0 means: photon has been detected
                 theStatus = Detection;
                 aParticleChange.ProposeLocalEnergyDeposit(thePhotonMomentum);
              }
              else {
                 aParticleChange.ProposeLocalEnergyDeposit(0.0);
              }

              NewMomentum = OldMomentum;
              NewPolarization = OldPolarization;

//              aParticleChange.ProposeEnergy(0.0);
              aParticleChange.ProposeTrackStatus(fStopAndKill);
}
```

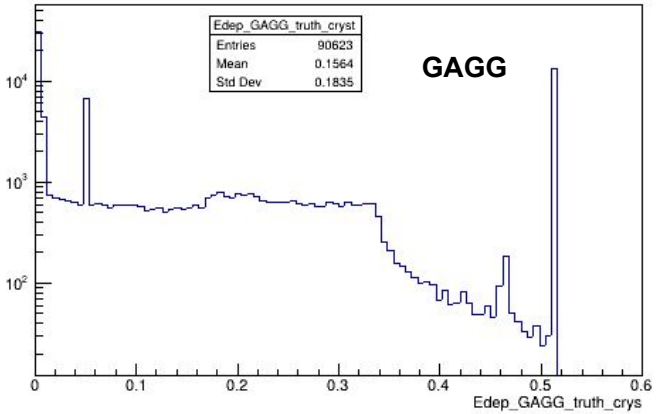
```
              if (PropertyPointer) {
                      theEfficiency =
                      PropertyPointer->Value(thePhotonMomentum);
              }
```

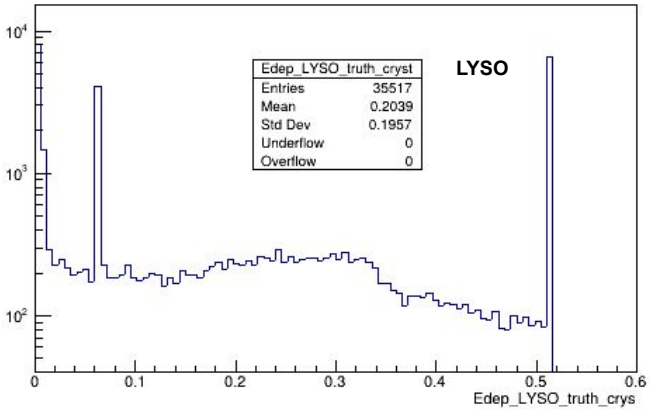
G4OpBoundaryProcess.cc

```
// Inline methods

inline
G4bool G4OpBoundaryProcess::G4BooleanRand(const G4double prob) const
{
  /* Returns a random boolean variable with the specified probability */

  return (G4UniformRand() < prob);
}

inline
```

G4OpBoundaryProcess.hh

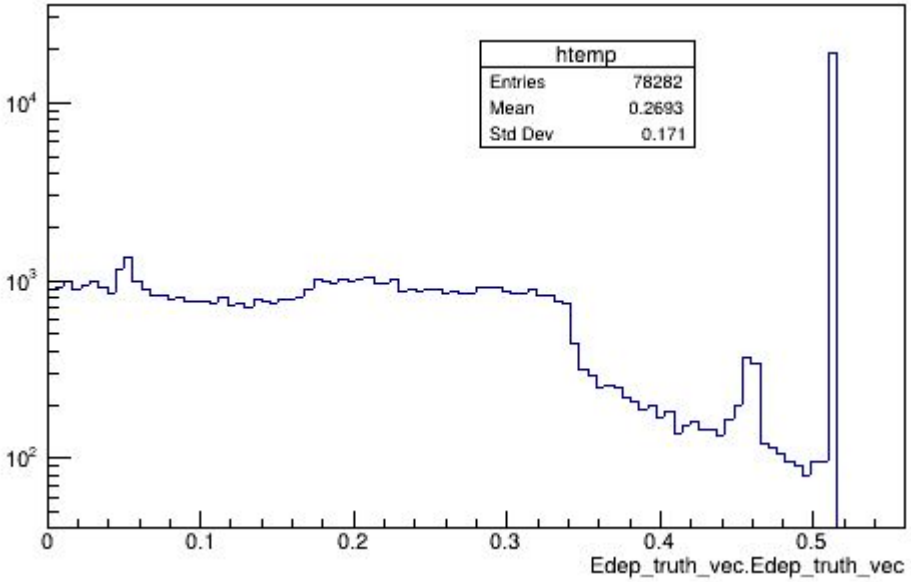
True Edep distribution for GAGG and LYSO individually





- This two individual distributions confirms that two spikes at 50 and 70 keV correspond to the BE of GAGG and LYSO.
- But it is not clear that why there isn't any spike near 430 keV in LYSO as it is in GAGG.
- Height of spike at 70 keV in LYSO is quite more than compared to 50 keV in GAGG.

True Edep distribution with different LYSO composition



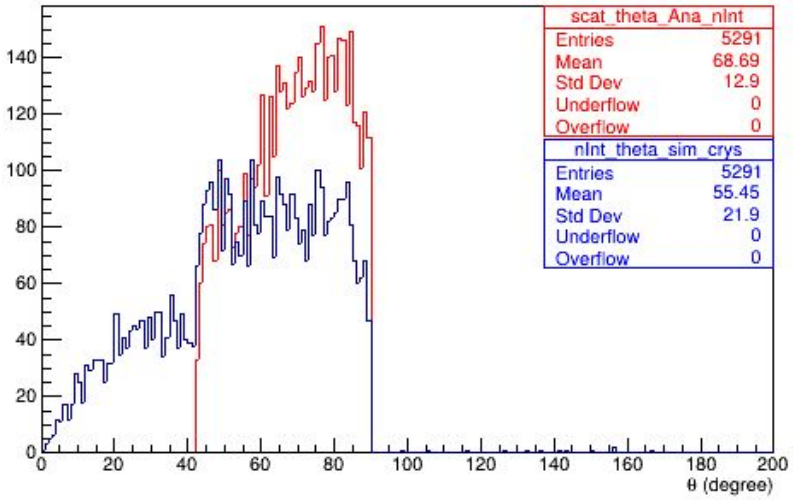
$LuYSiO_5$

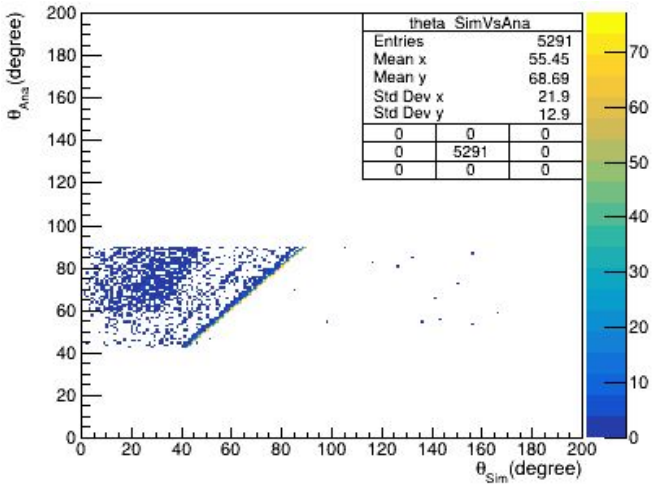
Code snippet

```
if(Hit_clrIdx[0] == 1 && Hit_clrIdx[1] ==2){
  //hypothesis 1: 0 is scatterer
  if (Hit_Edep[0] <= 0.255){    //forward scattering
    scat_idx = 0;
    abs_idx = 1;
  }

  //hypothesis 2: 1 is scatterer
  else if (Hit_Edep[1] > 0.255 && Hit_Edep[0] <= 0.34067){   //back scattering
    scat_idx = 1;
    abs_idx = 0;
  }
}

else if(Hit_clrIdx[0] == 2 && Hit_clrIdx[1] ==1){
  //hypothesis 1: 0 is scatterer
  if (Hit_Edep[0] > 0.255 && Hit_Edep[0] <= 0.34067){    //back scattering
    scat_idx = 0;
    abs_idx = 1;
  }

  //hypothesis 2: 1 is scatterer
  else if (Hit_Edep[1] <= 0.255){  //forward scattering
    scat_idx = 1;
    abs_idx = 0;
  }
}
```

```
if (scat_idx >=0){
  scat_Edep = Hit_Edep[scat_idx];

  if(scat_Edep < 0.34067){
  scat_theta = GetScatAngle(scat_Edep);

  h_theta->Fill(Deg(scat_theta));
  }
}
```

Comparing the nInt inclusive for valid theta_sim

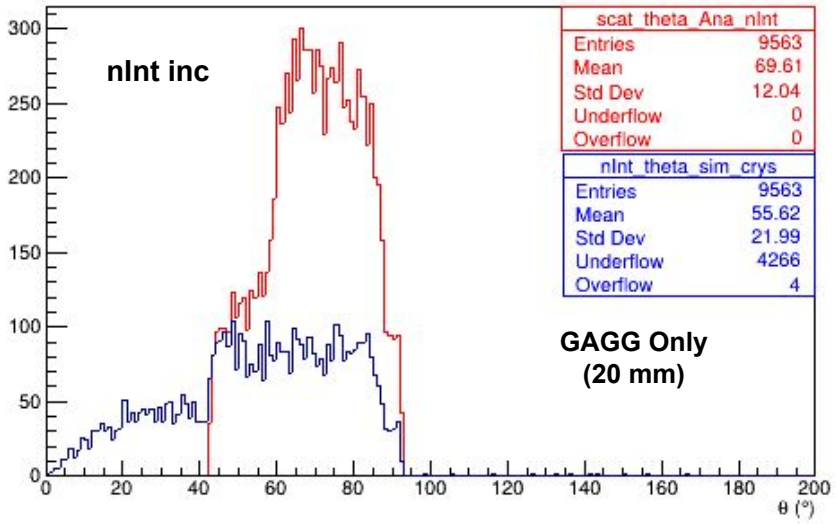


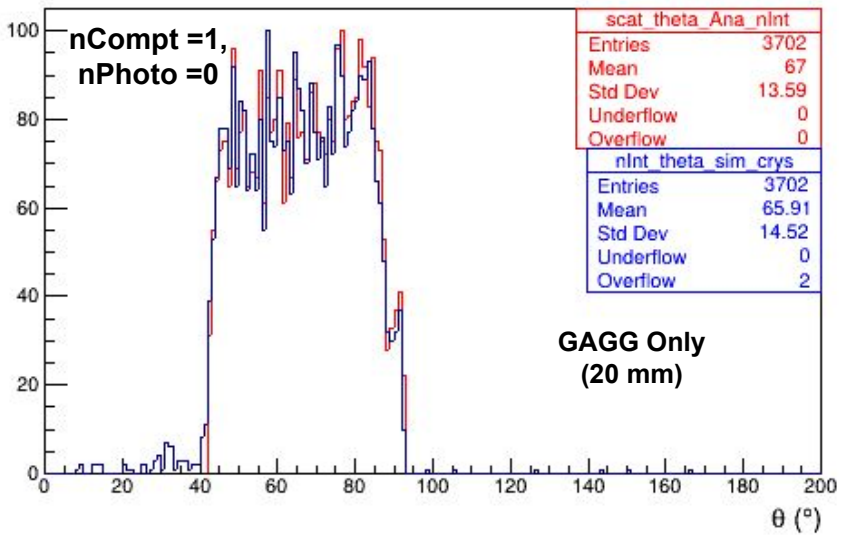



```
if(scat_theta >=0 && Hit_scat_theta_sim[scat_idx] >= 0){
  h_theta_ana_nInt->Fill(Deg(scat_theta));
  h_theta_sim[nInt]->Fill(Deg(Hit_scat_theta_sim[scat_idx]));
  h_theta_SimVsAna->Fill(Deg(Hit_scat_theta_sim[scat_idx]), Deg(scat_theta));

  double dTheta = Hit_scat_theta_sim[scat_idx]-scat_theta;
  h_dTheta->Fill(Deg(dTheta));
}
```

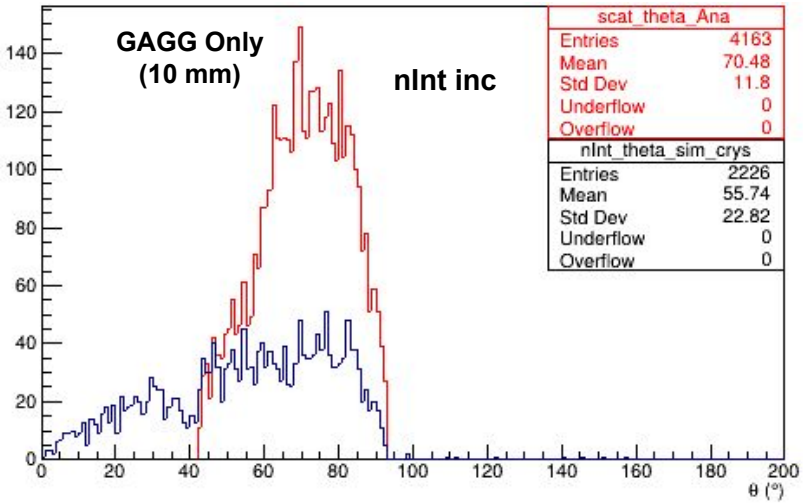
Date: 20/03/26

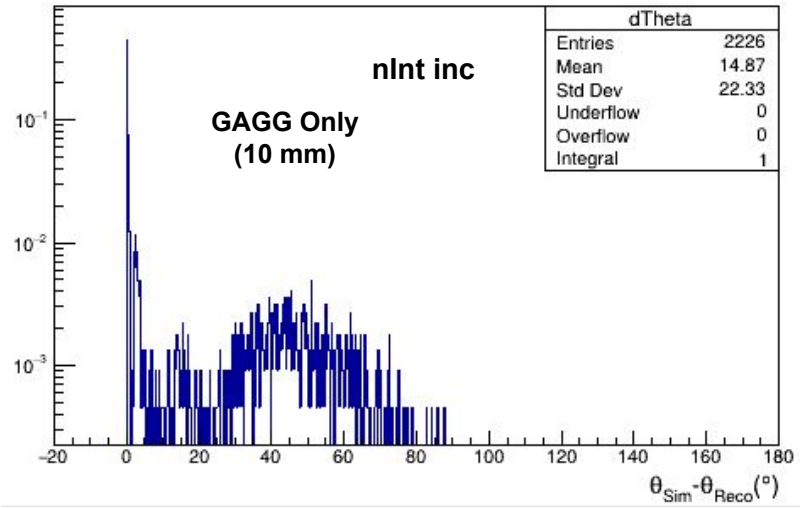
Checking after applying forwarding scattering condition on scatterer Edep for single crystal case



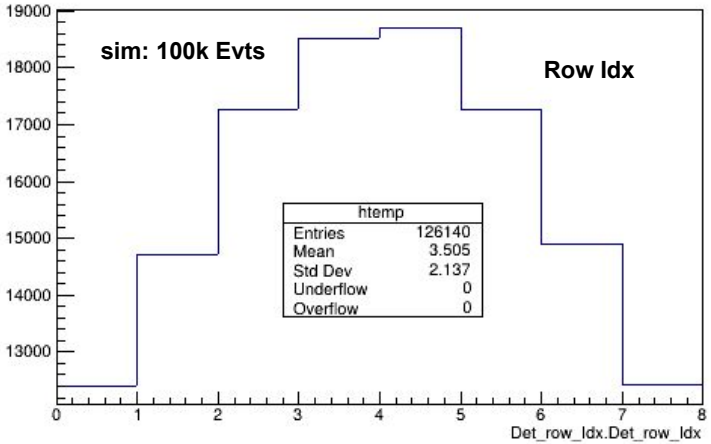


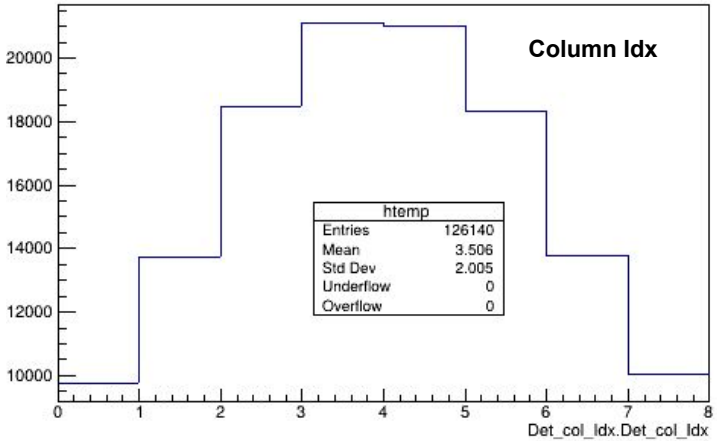
Checking with the lesser depth of GAGG (10 cm)

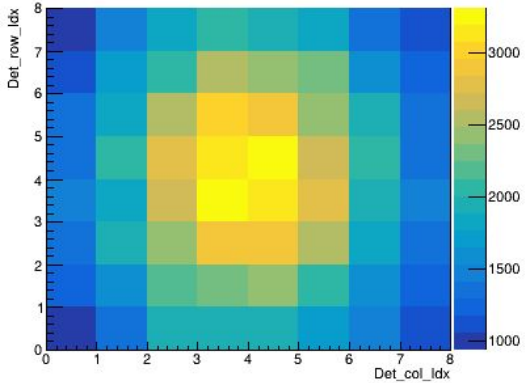


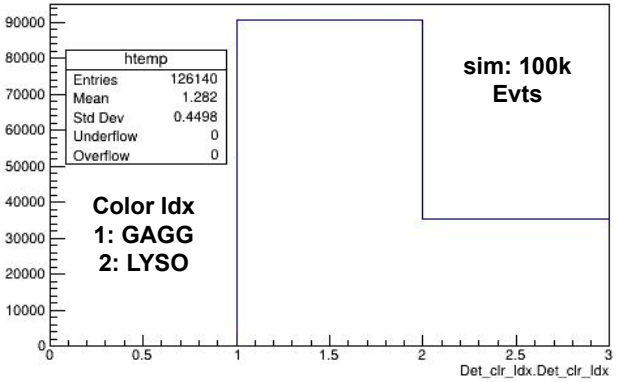


Distribution of positions of interactions



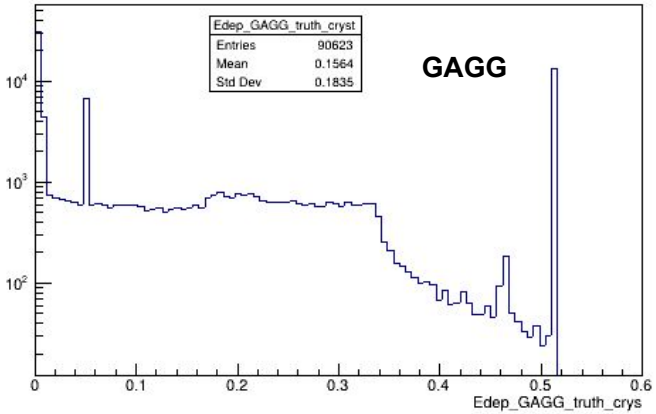


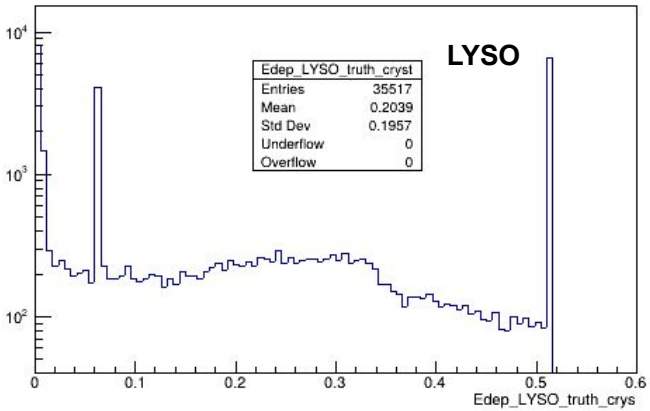




- Plotting the indices of interaction.
- We see more entries in the center region, because going away from the center changes the angle of incidence hence the interaction probability

True Edep distribution for GAGG and LYSO individually





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